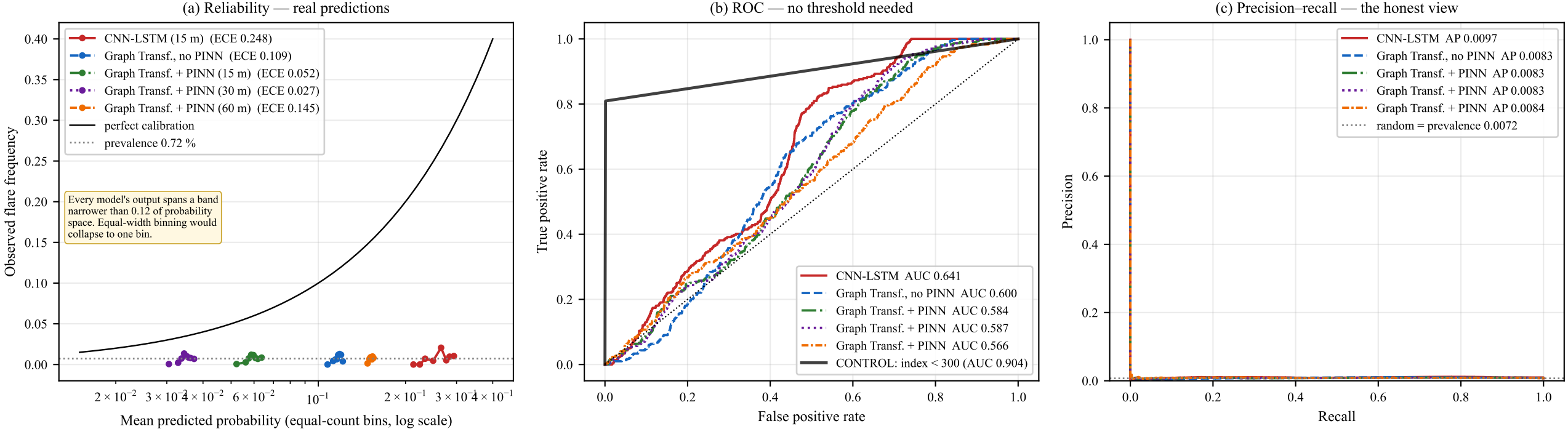

(a) Reliability — real predictions
CNN-LSTM (15 m) (ECE 0.248)
Graph Transf., no PINN (ECE 0.109)
Graph Transf. + PINN (15 m) (ECE 0.052)
Graph Transf. + PINN (30 m) (ECE 0.027)
Graph Transf. + PINN (60 m) (ECE 0.145)
perfect calibration
prevalence 0.72 %
Every model's output spans a band narrower than 0.12 of probability space. Equal-width binning would collapse to one bin.
Observed flare frequency
Mean predicted probability (equal-count bins, log scale)
(b) ROC — no threshold needed
CNN-LSTM AUC 0.641
Graph Transf., no PINN AUC 0.600
Graph Transf. + PINN AUC 0.584
Graph Transf. + PINN AUC 0.587
Graph Transf. + PINN AUC 0.566
CONTROL: index < 300 (AUC 0.904)
True positive rate
False positive rate
(c) Precision–recall — the honest view
CNN-LSTM AP 0.0097
Graph Transf., no PINN AP 0.0083
Graph Transf. + PINN AP 0.0083
Graph Transf. + PINN AP 0.0083
Graph Transf. + PINN AP 0.0084
random = prevalence 0.0072
Precision
Recall